# SPN Differential Lab - Study Guide & Learning Roadmap

Project: spn-dc (IT Security - Differential Cryptanalysis)
Goal: Pass the subject and finish the project without a deep dive into research-level cryptanalysis.

## 1. Topics covered by this project

### A. Crypto basics (minimum)

1. Block ciphers vs stream ciphers
2. Confusion and diffusion (Shannon)
3. Rounds, state, block size
4. XOR as "difference" in binary ciphers
5. Why we track differences, not full plaintext/ciphertext

### B. SPN structure (core)

6. Substitution-Permutation Network (SPN) layout
7. Parallel S-boxes (splitting the block into slices)
8. S-box as a permutation table (lookup, bijective)
9. Linear / permutation layer after S-boxes (P-layer)
10. One round = S-box layer -> P-layer (keys often ignored in difference analysis)
11. Multi-round repetition

### C. Differential cryptanalysis (exam-level)

12. Input difference Deltax and output difference Deltay
13. Differential trail / characteristic (path through rounds)
14. DDT (Difference Distribution Table)
15. DDT formula: count pairs where S(x XOR a) XOR S(x) = b
16. DDT probability ? count / 2^n for an n-bit S-box
17. Forward propagation (input diff -> possible output diffs)
18. Backward / inverse view (fixed output -> possible inputs)
19. "Strong" vs "weak" differentials (high vs low DDT counts)

### D. Linear layer math (just enough)

20. GF(2) / binary arithmetic (add = XOR)
21. Binary matrix as bit mixer
22. Bijective = invertible = full rank (no collapsing)
23. Differences through P: same linear map as data (for XOR differences)

### E. Constraints (project features)

24. Fixed difference at a position
25. Abstract / nonzero-unknown difference
26. Equality constraints (two S-box inputs must match)
27. Conflict detection (empty possibility set = impossible)

### F. Software / project (finish & demo)

28. Python package layout (spn_dc/)
29. CLI commands (spn-dc, session file)
30. Streamlit UI (tabs, propagate, animation, bit diagram)
31. Running tests (pytest)
32. Short report / demo for submission

## 2. Learning roadmap (~1-2 weeks, shallow)

Objective: Understand enough to explain the project, run a demo, and answer viva questions.

### Phase 1 - Big picture (Day 1-2, ~3 hours)

Learn | Stop when you can...
Block cipher + SPN diagram | Draw one round on paper: S-boxes -> P
XOR difference | Say "difference = XOR of two related inputs"
What DDT is for | Say "DDT tells which output diffs are likely for a given input diff"

Skip: full AES, Feistel details, key schedules, real attacks on ciphers.
Read: PROJECT_REPORT.md sections 1-3.

### Phase 2 - S-box + DDT (Day 3-4, ~4 hours)

Learn | Stop when you can...
S-box as permutation table | Give an example 4-bit table
Build / read one DDT row | For input diff a, list possible b with count > 0
DDT probability | Compute prob = count/16 for 4-bit box

Practice: UI -> DDT tab; CLI: compute_ddt --sbox 0.

**Exam phrases:**

- Higher DDT count -> more likely differential
- Row for Deltax = 0 is always concentrated at Deltay = 0

### Phase 3 - P-layer + bijective matrix (Day 5, ~2 hours)

Learn | Stop when you can...
Binary matrix over GF(2) | "Each output bit is XOR of some input bits"
Bijective | "Invertible; every output pattern has one preimage"
Differences through P | "P acts linearly on differences the same way as on data"

Practice: UI -> S-boxes & P; Bit wiring tab for pins and P connections.
Skip: Gaussian elimination proofs, optimal P design.

### Phase 4 - Propagation & constraints (Day 6-7, ~3 hours)

Learn | Stop when you can...
One-round flow | before S-box -> after S-box -> after P
Multi-round | Output of P becomes next round's S-box input

Constraints | Fixed / abstract / equal; check = consistent or not
Explosion of possibilities | Why small examples only

**Practice:**

1. Create cipher: 2 rounds, 2 S-boxes, 4-bit
2. Propagate 0x11
3. Play animation + Bit wiring
4. Add conflicting equality -> Check constraints fails

### Phase 5 - Trails & paths (Day 8, ~2 hours)

Learn | Stop when you can...
Differential trail | One choice of output diff per S-box per round
Trail probability | Product of per-S-box DDT probabilities
generate_paths / filter | List likely trails above threshold

Practice: Paths tab -> generate -> filter -> animate one trail.

### Phase 6 - Finish for submission (Day 9-10, ~4 hours)

Task | Deliverable
Pick one toy cipher | e.g. 2x4-bit S-boxes, 3 rounds
One worked example | Input Delta, 1-2 rounds propagation (screenshot)
One DDT screenshot | Comment on one "strong" cell
One constraint example | Equal or fixed diff that works / fails
Short write-up (1-2 pages) | Problem, design, DDT, propagation, demo, limits
Optional | pytest green + short demo video

## 3. Viva cheat sheet

1. SPN: parallel S-boxes, then linear bit mixing, repeated for R rounds.
2. Difference: XOR; we study how Delta propagates; keys simplified away.
3. DDT: DDT[a][b] = number of x with S(xXORa)XORS(x) = b.
4. Bijective P: invertible binary matrix; differences follow same map.
5. This tool: configures SPN, DDT, propagates Delta, constraints, optional paths.
6. Limitation: exponential blow-up; educational sizes only; not a full attack tool.

## 4. Must know vs can ignore

Must know (pass) | Can ignore (for now)
SPN round structure | Feistel, ARX ciphers
XOR difference | Modular addition differentials
DDT meaning + one example | Optimal S-box design theory
Binary matrix P-layer | MDS matrices, coding theory
Bijective = invertible | Formal attack proofs
CLI/UI demo flow | Attacks on real AES
Project limitations | Key schedule proofs

## 5. Fastest study order

```
Day 1-2:  SPN + XOR differences
Day 3-4:  S-box + DDT (UI DDT tab)
Day 5:    P-layer + bijective matrix (Bit wiring tab)
Day 6-7:  Propagation + constraints
Day 8:    Paths (good for report)
Day 9-10: Write report + rehearse demo
```

## 6. Five-minute demo script

1. Sidebar: 2 rounds, 2 S-boxes, 4-bit -> Create cipher
2. DDT tab: show S-box 0; mention one high-probability entry
3. Propagate 0x11 -> Play animation
4. Bit wiring: scrub steps; point at S0, S1, P wires
5. Input & constraints: fix one input diff -> propagate again
6. Paths: generate trails, filter, show strongest trail
7. Close: "Tool models difference propagation; real ciphers need larger analysis and key handling."

## 7. Key equations (exam)

**Difference (XOR):**

```
Deltax = x XOR x'
```

**DDT entry:**

```
DDT[a][b] = |{ x : S(x XOR a) XOR S(x) = b }|
```

**DDT probability (n-bit S-box, 2^n inputs):**

```
P(b | a) ? DDT[a][b] / 2^n
```

**P-layer (linear over GF(2)):**

```
y = M . x   (addition = XOR, multiplication = AND)
```

Bijective P: matrix M is invertible (full rank over GF(2)).

*Generated for the spn-dc project - differential propagation lab for configurable SPNs.*